CHAPTER 1

Tang & Rong

Rong Tang

2019–2025

An- nals of Sta- tis- tics

MCMC

1.

1.1.

•

D

$d \ll$
D

D

GAN
VAE

(1)

(2)

(3)

MCMC

Tang-
Yang

.

Tang & Yang 2019–2025

8

MCMC

1

2

VAE

3

MCMC

2.

2.1.

.

$\mathcal{M}^*$

$\mathbb{R}^D$

d

β-

$\beta \geq$
1

P^*

$\mathcal{M}^*$

$\mathrm{vol}_{\mathcal{M}^*}$

p^*

α-

$\alpha \in$
$[0, \beta -$
$1]$

p^*

$\mathbb{R}^D$

Lebesgue

Lebesgue

Partition of Unity Tang

$(U_\lambda, \varphi_\lambda)$

$\{\gamma_\lambda\}$

Tang
(2022)

Tang

(2023a)

Reach

reach

---

reach($\mathcal{M}$) =
inf{r >
0 :
$\mathcal{M}$

r-

}

2.2.

IPM.

KL

Hellinger

Tang

IPM

$$d_{\mathcal{F}}(\mu,\nu)=\sup_{f\in\mathcal{F}}\Big|\int_{\mathbb{R}^D} f(x)\,\mathrm{d}\mu-\int_{\mathbb{R}^D} f(x)\,\mathrm{d}\nu\Big|.$$

$\mathcal{F}$

γ-
Hölder

Sobolev

1-
Lipschitz

Wasserstein-
1

RKHS

MMD
γ

trade-
off

$\gamma =$
1

d_γ

Wasserstein-
1

$\gamma \to$
0^+

d_γ

Tang
(2025a)

2.3.

.
Tang

$$= n^{-1/2} \vee n^{-\frac{\alpha+\gamma}{2\alpha+d}} \vee n^{-\frac{\beta\gamma}{d}}.$$

$n^{-1/2}$

d

β-

d

Haus-
dorff

D

Lep-
ski

chain-
ing
peel-
ing
lo-
cal-
iza-
tion

Gir-
sanov

Poincaré

2.4.

.
**Gauss-
ian**

Langevin

$p_\sigma(x) = \int p_{\mathrm{data}}(y)\phi_\sigma(x - y)\,\mathrm{d}y$

L_∞

-

$\nabla \log p_t(x)$

t

$t =$
0

$t =$
T

Tang
(2024a)

$\mathcal{S}_{\mathrm{NN}}$

t

Girsanov

Gir-
sanov

SDE

3.

3.1. Tang & Yang (2022) —

.

Rong Tang and Yun Yang. "Minimax Rate of Distribution Estimation on Unknown Submanifold under Adversarial Losses." *Annals of Statistics*, 2022.

D

GAN
VAE

γ

$$n^{-1/2} \vee n^{-\frac{\alpha+\gamma}{2\alpha+d}} \vee n^{-\frac{\beta\gamma}{d}},$$

α

β

d

D

par-
ti-
tion
of
unity

VAE

Tay-
lor

Lep-
ski

Fano

Varshamov-
Gilbert

Tang
(2022)

Tang
(2023a,
2024a)

3.2. Tang & Yang (2023a) —

•

Rong Tang and Yun Yang. “Estimating Distributions with Low-dimensional Structures Using Mixtures of Generative Models.” *ICML*, 2023.

VAE
GAN

MLD-
MAE

K

-

ρ_k

1-
Wasserstein

α-

β-

d

$$O\left(n^{-\frac{\alpha\wedge(\beta-1)+1}{2(\alpha\wedge(\beta-1))+d}}\right) \vee O\left(\frac{\log n}{\sqrt{n}}\right),$$

$\alpha \leq$
$\beta -$
1

/

Tang (2022)

Tang (2022)

3.3. Tang et al. (2023b) —

•

Rong Tang, Anirban Bhattacharya, Debdeep Pati, and Yun Yang. "Robust Bayesian Inference on Riemannian Submanifold," *JRSS B*, 2023.

Stiefel

Un-
cer-
tainty
Quan-
tifi-
ca-
tion

Bayesian

RPE-
TEL

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(1)

(2)

(3)

Bernstein-
von
Mises
(BvM)

set-
ting

**Me-
trop-
o-
lis
(RRWM)**

d

D

conductance profile

3.4. Tang & Yang (2023c) —

•

Rong Tang and Yun Yang. “Minimax Nonparametric Two-Sample Test under Adversarial Losses,” *AISTATS*, 2023.

MMD

$O(n^2)$

Besov

Besov

1

ℓ_1
ℓ_2

Wasser-
stein

1

2

$O(n \log n +$

$n^{2d/(4\alpha+d)})$

MMD

$O(n^2)$

3.5. Tang & Yang (2024a) —

.

Rong Tang and Yun Yang. "Adaptivity of Diffusion Models to Manifold

Struc-
tures,"
*AIS-
TATS*,
2024.

SOTA

Langevin

–

d

D

Langevin

Gauss-
ian

$\tilde{O}(\sigma)$

σ

n

d

L_∞

$-$

1-
Wasserstein

$$\frac{1}{\sqrt{n}} \vee n^{-\frac{\alpha+1}{2\alpha+d}}.$$

Tang
(2022)

Tang
(2022)

3.6. Tang (2024b) — Metropolis-Adjusted Langevin

.

Rong Tang and Yun Yang. "Metropolis-Adjusted Langevin Algorithm: Optimal Dimension Dependency and Beyond." *ICML*, 2024.

MALA

MCMC

$O(d^{1/3})$

**s-
conductance
pro-
file**

MALA

$O(d^{1/3})$

burn-
in

s-
conductance
pro-
file

con-
duc-
tance

Chen
et
al.
(2020)

con-
duc-
tance
pro-
file

warm-
start

$\log M_0$

$\log \log M_0$

Tang (2023b)

Tang (2023b)

BvM

BvM

MALA

3.7. Tang & Yang (2025a) —

.

Rong Tang and Yun

Yang. "Minimax Optimal Rates for Regression on Manifolds and Distributions," arXiv, 2025.

X

Y

$\mu^*_{Y|X}$

X

Y

IPM

(1)
Regime 1

Lebesgue

$$n^{-\frac{\alpha_X}{2\alpha_X+d_X}}+n^{-\frac{\alpha_Y+\gamma}{2\alpha_Y+D_Y+\frac{\alpha_Y}{\alpha_X}\cdot d_X}}.$$

(2)
Regime 2

:

$\mathcal{M}_{Y|X}$

X

(3) $n^{-\gamma\beta_Y/d_Y}$

Regime 3

:

$\mathcal{M}_{Y|x}$

x

$n^{-\gamma/(\frac{d_Y}{\beta_Y}+\frac{d_X}{\beta_X})}$

γ-
Hölder
IPM
d_γ

$\gamma =$
0

1-
Wasserstein

$\gamma =$
1

3.8. Tang & Yang (2025b) —

.

Rong Tang and Yun Yang. “Conditional Diffusion Models are Minimax-Optimal and Manifold-Adaptive,” arXiv, 2025.

Tang
(2024a)

(1)

$\alpha_X \in$
$[0, 1]$

(2)

X

Y

Wasser-
stein

(d_X, d_Y)

(D_X, D_Y)

lo-
cal-
iza-
tion
and
peel-
ing
tech-
niques

Gir-
sanov

Tang

(2024a, 2025a)

Tang (2024a)

→
Tang (2025a)

→

4.

Tang-
Yang

2019–
2025

4.1.

.

Tang (2022) $\longrightarrow$ Tang (2023a) MLDMAE $\longrightarrow$ Tang (2025a)

4.2.

VAE

.

VAE (Tang 2021) $\longrightarrow$ Tang (2024a) $\longrightarrow$ Tang (2025b)

Tang
(2024a)

$t >$
0

Lebesgue

4.3.

MCMC.

Tang (2023b) BvM + RRWM $\xrightarrow{\text{BvM}}$ Tang (2024b) MALA

BvM

MALA

$d^{1/3}$

5.

5.1.

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(1)
2021
—
Tang
&
Yang,
COLT
VAE

(2)
2022
—
Tang
&
Yang,
An-
nals
of
Sta-
tis-
tics

(3) **2023a** — Tang & Yang, ICML MLD-MAE

(4) **2023b** — Tang et al., JRSS B

+ BvM

(5) **2023c** — Tang & Yang, AIS-TATS

(6)
2024a
—
Tang
&
Yang,
AIS-
TATS

(7)
2024b
—
Tang
&
Yang,
ICML
MALA

(8)
2025a
—
Tang
&
Yang,
arXiv

(9)
2025b
—
Tang
&
Yang,
arXiv

5.2.

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(1)

**Par-
ti-
tion
of
Unity**
Tang
(2022,
2023a)

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(2)
γ-
Hölder
IPM
Tang
(2022,
2023c,
2025a)

(3)
Gaussian

+

Tang
(2024a)

Langevin

L_∞

(4)

Tang
(2024a,
2025b)

t

t

(5)

s-**conductance profile** Tang (2024b)

conductance

con-
duc-
tance
pro-
file

(6)

**Bernstein-
von
Mises**

Tang
et
al.
(2023b)

BvM

set-
ting

(7)
Girsanov

Tang
(2024a,
2025b)

Gir-
sanov

SDE

6.

Tang-
Yang

(1)

d

d

d

(2)

Tang
(2022)

(3)

Tang
(2023c)

J

α

α

(4)

(5)

ResNet
U-
Net

Tang-
Yang

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7.

Bibliography

[1] Tang, R. and Yang, Y. (2021). Excess risk bounds for distribution estimation with VAE. *COLT*, 2021.

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AISTATS, 2023.

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